

Conditional probability

$P(A | B) = P(A \text{ and } B) / P(B)$, provided $P(B) > 0$.

The conditioning event changes the sample space.

Example: $P(A \text{ and } B) = 0.12$ and $P(B) = 0.30$.

Then $P(A | B) = 0.40$.

Bayes' theorem

$P(A | B) = P(B | A) P(A) / P(B)$, provided $P(B) > 0$.

A prior probability is updated after observing evidence.

This authored evaluation sheet is not an assessment.

Independence

Events A and B are independent when $P(A \text{ and } B) = P(A) P(B)$.

If $P(B) > 0$, independence implies $P(A | B) = P(A)$.

Do not confuse independence with mutually exclusive events.